Special Relativity Notes

Role of Special Relativity in Electricity and Magnetism

2/11/26

1 Role of Special Relativity in Electricity and Magnetism

Electricity and magnetism are consistent with the special theory of relativity.

Objective: to show that Maxwell's equations are consistent with the special theory of relativity ($\vec{E}$, $\vec{B}$ fields).

First, we need to find the transformation equations of the $\vec{E}$ and $\vec{B}$ fields, i.e. how

$$E_x,\ E_y,\ E_z \qquad \text{and} \qquad B_x,\ B_y,\ B_z$$

transform under the Lorentz transformation.

2 Thought Experiment

A charge q is moving with a certain velocity $\Rightarrow$ it produces a $\vec{B}$ field.

A test charge q_t moves with velocity $\vec{v}$. The magnetic force on it, in frame S, is

$$\vec{F}_m = q_t\,(\vec{v} \times \vec{B}) \qquad \text{(in frame } S\text{)}$$

S'-frame (moving with velocity $\vec{v}$)

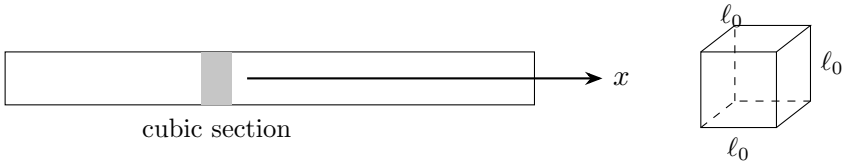
$$\vec{F}'_m = 0$$

This happens when

- the observer sits on the **test charge** ($\vec{v}' = 0$), or
- the observer sits on the **source charge** ($\vec{B} = 0$).

3 Charge Density of a Current-Carrying Wire

Consider a wire along the x-axis and a small cubic section of side ℓ_0.



In the rest frame:

- volume $= \ell_0^3$
- N = number of electrons, so the total charge is $Q = Ne$

$$\text{charge density (at rest):} \qquad \rho_0 = \frac{Ne}{\ell_0^3}$$

Apply a potential difference $\Rightarrow$ the electrons move with velocity $\vec{u}$.

Length contracts along the x-axis to $\ell_0\sqrt{1-\dfrac{u^2}{c^2}}$, so

$$\text{new volume} = \ell_0^3\sqrt{1-\frac{u^2}{c^2}}$$

$$\text{charge density:} \qquad \rho = \frac{Ne}{\ell_0^3\sqrt{1-\dfrac{u^2}{c^2}}} = \frac{\rho_0}{\sqrt{1-\dfrac{u^2}{c^2}}}$$

4 Current Density and the Four-Current

Current density = current per unit area:

$$\vec{J} = \rho\,\vec{u}$$

For arbitrary $\vec{u}$, $\vec{J}$ has components J_x, J_y, J_z:

$$J_x = \rho u_x, \qquad J_y = \rho u_y, \qquad J_z = \rho u_z$$

Here only $J_x \neq 0$, since the motion is along the x-axis ($J_y = J_z = 0$).

$\vec{J} \longleftrightarrow \rho$ — these two are to be combined into one four-vector, just like

$$x^\mu = (ct, \vec{x}), \quad x^\mu x_\mu = c^2t^2 - \vec{x}^{\,2} \qquad\qquad p^\mu = \left(\frac{E}{c}, \vec{p}\right), \quad p^\mu p_\mu = \frac{E^2}{c^2} - \vec{p}^{\,2}$$

$$J^\mu = (\rho c,\ \vec{J}), \qquad J^\mu J_\mu = \rho^2 c^2 - \vec{J}^{\,2}$$

5 How Do $\vec{J}$ and ρ Transform under the Lorentz Transformation?

From now on, let v be the speed of the charge carriers in the frame S, and u the speed of the frame S' relative to S.

ρ transforms like the mass m

$$\rho = \frac{\rho_0}{\sqrt{1-\dfrac{v^2}{c^2}}} = \frac{\rho_0}{\sqrt{1-\dfrac{v^2}{c^2}}} \cdot \frac{m_0}{m_0} = \frac{\rho_0}{m_0}\, m$$

ρ transforms in the same way as m.

$\vec{J}$ transforms like the momentum $\vec{p}$

$$\vec{J} = \rho\,\vec{v} = \frac{\rho_0}{\sqrt{1-\dfrac{v^2}{c^2}}}\,\vec{v}\cdot\frac{m_0}{m_0} = \frac{\rho_0}{m_0}\,\vec{p}$$

where $\vec{p} = \dfrac{m_0\vec{v}}{\sqrt{1-\dfrac{v^2}{c^2}}}$ is the momentum.

$\vec{J}$ transforms in the same way as $\vec{p}$.

So it is enough to find how $(E, \vec{p})$ transform.

6 Exercise: Transformation of $\vec{p}$ and E

To be shown (analogous to x', y', z', t'):

$$p'_x = \frac{p_x - \dfrac{u}{c^2}E}{\sqrt{1-\dfrac{u^2}{c^2}}}, \qquad p'_y = p_y, \qquad p'_z = p_z, \qquad E' = \frac{E - u\,p_x}{\sqrt{1-\dfrac{u^2}{c^2}}}$$

In frame S:

$$\vec{p} = \frac{m_0\vec{v}}{\sqrt{1-\dfrac{v^2}{c^2}}}, \qquad p_x = \frac{m_0 v_x}{\sqrt{1-\dfrac{v^2}{c^2}}}, \quad p_y = \frac{m_0 v_y}{\sqrt{1-\dfrac{v^2}{c^2}}}, \quad p_z = \frac{m_0 v_z}{\sqrt{1-\dfrac{v^2}{c^2}}}, \quad E = \frac{m_0 c^2}{\sqrt{1-\dfrac{v^2}{c^2}}}$$

In frame S' (velocity addition): $\quad v'_x = \dfrac{v_x - u}{1-\dfrac{uv_x}{c^2}}, \quad v'_y = \dfrac{v_y\sqrt{1-\dfrac{u^2}{c^2}}}{1-\dfrac{uv_x}{c^2}}$, and

$$\frac{1}{\sqrt{1-\dfrac{v'^2}{c^2}}} = \frac{1}{\sqrt{1-\dfrac{u^2}{c^2}}}\cdot\frac{1-\dfrac{uv_x}{c^2}}{\sqrt{1-\dfrac{v^2}{c^2}}}$$

p'_x

$$p'_x = \frac{m_0\, v'_x}{\sqrt{1 - v'^2/c^2}} = m_0\, \frac{v_x - u}{1 - \dfrac{uv_x}{c^2}} \cdot \frac{1}{\sqrt{1 - \dfrac{u^2}{c^2}}} \cdot \frac{1 - \dfrac{uv_x}{c^2}}{\sqrt{1 - \dfrac{v^2}{c^2}}}$$

$$p'_x = \frac{m_0\,(v_x - u)}{\sqrt{1 - \dfrac{u^2}{c^2}}\sqrt{1 - \dfrac{v^2}{c^2}}}$$

$$= \frac{1}{\sqrt{1 - \dfrac{u^2}{c^2}}} \left(\underbrace{\frac{m_0 v_x}{\sqrt{1 - \dfrac{v^2}{c^2}}}}_{p_x} - \underbrace{\frac{m_0 u}{\sqrt{1 - \dfrac{v^2}{c^2}}}}_{uE/c^2} \right)$$

$$\boxed{p'_x = \frac{p_x - \dfrac{uE}{c^2}}{\sqrt{1 - \dfrac{u^2}{c^2}}}}$$

p'_y and p'_z

$$p'_y = \frac{m_0 v'_y}{\sqrt{1 - v'^2/c^2}} = \frac{m_0}{\sqrt{1 - v'^2/c^2}} \cdot \frac{v_y\sqrt{1 - \dfrac{u^2}{c^2}}}{1 - \dfrac{uv_x}{c^2}} = \frac{m_0 v_y}{\sqrt{1 - \dfrac{v^2}{c^2}}} = p_y$$

and similarly $p'_z = p_z$.

$$p'_y = p_y, \qquad p'_z = p_z$$

E'

$$E' = \frac{m_0 c^2}{\sqrt{1 - v'^2/c^2}} = \frac{m_0 c^2}{\sqrt{1 - \dfrac{u^2}{c^2}}\sqrt{1 - \dfrac{v^2}{c^2}}} \left(1 - \frac{uv_x}{c^2}\right)$$

$$= \frac{1}{\sqrt{1 - \dfrac{u^2}{c^2}}} \left(\underbrace{\frac{m_0 c^2}{\sqrt{1 - \dfrac{v^2}{c^2}}}}_{E} - \underbrace{\frac{m_0\, u\, v_x}{\sqrt{1 - \dfrac{v^2}{c^2}}}}_{u\, p_x} \right)$$

$$E' = \frac{E - u\,p_x}{\sqrt{1 - \dfrac{u^2}{c^2}}}$$

7 Transformation of $\vec{J}$ and ρ

Since $\vec{J} = \dfrac{\rho_0}{m_0}\vec{p}$ and $\rho = \dfrac{\rho_0}{m_0}m = \dfrac{\rho_0}{m_0}\dfrac{E}{c^2}$, multiplying the transformation of $(\vec{p}, E)$ by ρ_0/m_0 gives:

$J'_x = \dfrac{J_x - u\rho}{\sqrt{1 - \dfrac{u^2}{c^2}}}$	$J'_y = J_y$	$J'_z = J_z$	$\rho' = \dfrac{\rho - \dfrac{uJ_x}{c^2}}{\sqrt{1 - \dfrac{u^2}{c^2}}}$